



尼日利亚

5MMSCFD 天然气液化装置

Technical Proposal for 5MMSCFD LNG Plant in
Nigeria

四川金星清洁能源装备集团股份有限公司

Sichuan Jinxing Clean Energy Equipment Co., Ltd.

2026 年 3 月

第 1 章 装置设计基础条件	CHAPTER 1 BASIC DESIGN CONDITIONS OF THE PLANT	4
1.1 项目概况	PROJECT OVERVIEW	4
1.2 计量单位	UNITS OF MEASUREMENT	4
1.3 原料气	FEED GAS	5
1.4 建设地自然条件	NATURAL CONDITIONS OF THE CONSTRUCTION SITE	5
1.5 操作弹性和设计寿命	OPERATIONAL FLEXIBILITY AND DESIGN LIFE	6
1.6 可利用的公用工程条件	AVAILABLE UTILITY CONDITIONS	6
1.6.1 电源	Power Supply	6
第 2 章 产品方案和公用工程消耗	CHAPTER 2 PRODUCT SCHEME AND UTILITY CONSUMPTION	7
2.1 生产规模和产品方案	PRODUCTION SCALE AND PRODUCT SCHEME	7
2.1.1 产品方案	Product Proposal	7
2.2 公用工程消耗	UTILITY CONSUMPTION	8
2.2.1 装置电耗	Plant Power Consumption	8
2.2.2 装置冷却方式	Plant Cooling Method	9
2.2.3 燃料气消耗	Fuel Gas Consumption	10
2.2.4 装置其他消耗	Other Plant Consumption	10
2.3 三废排放	WASTE DISCHARGE	11
第 3 章 工艺技术方案及说明	CHAPTER 3 PROCESS TECHNICAL SCHEME AND DESCRIPTION	12
3.1 系统概述	SYSTEM OVERVIEW	12
3.1.1 设计方案概述	Design Scheme Overview	12

3.1.2 供货范围概述 Supply Scope Overview	14
3.2 各系统说明 DESCRIPTION OF EACH SYSTEM	15
3.2.1 原料气调压、过滤分离、增压、计量系统 Feed Gas Pressure Regulation, Filtration & Separation, Pressurization and Metering System	15
3.2.2 原料气脱酸气单元 Feed Gas Acid Removal Unit	16
3.2.3 原料气干燥及脱汞单元 Feed Gas Drying and Mercury Removal Unit	18
3.2.4 液化单元 Liquefaction Unit	21
3.2.5 制冷单元 Refrigeration Unit	22
3.2.6 冷剂储存、配比及回收单元 Refrigerant Storage, Proportioning and Recovery Unit	23
3.2.7 系统干燥和解冻单元 System Drying and Thawing Unit	25
3.2.8 产品储存及装车单元 Product Storage and Loading Unit	26
3.2.9 BOG 回收单元 BOG Recovery Unit	26
3.2.10 污水储存单元 Waste Water Storage Unit	27
3.2.11 燃料气单元 Fuel Gas Unit	27
3.2.12 安全排放单元 Safety Venting Unit	28
第 4 章 设备选型和供货 CHAPTER 4 EQUIPMENT SELECTION AND SUPPLY	29
4.1 工艺设备 PROCESS EQUIPMENT	29

第 1 章 装置设计基础条件

Chapter 1 Basic Design Conditions of the Plant

1.1 项目概况|Project Overview

本天然气液化项目概况如表 1.1-1。
 The overview of this natural gas liquefaction project is presented in Table 1.1-1.

表 1.1-1 项目概况
 Table 1.1-1 Project Overview

项目名称 Project Name	撬装式天然气液化装置 Skid-mounted Natural Gas Liquefaction Plant
建设地点 Construction Site	尼日利亚 Nigeria
原料气 Feed Gas	天然气 Natural Gas
建设规模 Construction Scale	天然气处理量按 5MMSCFD；年处理量年操作时间按 8000 小时。 Natural gas processing capacity: 5MMSCFD; Annual operating hours: 8000 hours
工艺流程 Process Flow	本项目采用 MRC 制冷工艺。 MRC refrigeration process adopted for this project

1.2 计量单位|Units of Measurement

除非另有说明，本技术文件的将按如下的国家法定计量单位：

Unless otherwise specified, the national statutory units of measurement listed below shall apply to this technical document:

温度 Temperature	°C
压力 Pressure	barg
流量 Flow rate	Nm ³ /h (referring to gas state at 0°C, 0.101325MPa.A)
流量 Flow rate	MMSCFD
功率 Power	kW
组成 Composition	mol%

1.3 原料气|Feed Gas

原料气的状态参数和组分，如表 1.3-1。

The state parameters and composition of the feed gas are shown in Table 1.3-1.

表 1.3-1 原料气状态条件

Table 1.3-1 Feed Gas State Conditions

原料天然气日处理量 Daily Feed Gas Processing Capacity		5MMSCFD	
原料气进厂压力 Inlet Pressure of Feed Gas to the Plant		402.1psi(28bar)~522.13 psi(36bar)	
原料气进厂温度 Inlet Temperature of Feed Gas to the Plant		38℃	
原料气组成 Feed Gas Composition (mol%)			
参数 Component		参考数据 Value	备注 Remarks
	Nitrogen	0.008	
	CO2	0.007	设计点：3% Design point: 3%
	H2S	0	
	Methane	0.900	
	Ethane	0.050	
	Propane	0.025	
	Ibutane	0.010	
	汞 Mercury	未检测 Not detected	

说明：

- 以上组成数据由用户提供。

Note:

- The above composition data is provided by the Client.

1.4 建设地自然条件|Natural Conditions of the Construction Site

环境温度|Ambient temperature：24°C – 36°C

1.5 操作弹性和设计寿命|Operational Flexibility and Design Life

装置操作弹性|Plant operational flexibility: 50 ~ 110%。

动设备平均无故障运行时间|Mean time between failures (MTBF) of rotating equipment: 8000h。

静设备设计寿命|Design life of static equipment: 20 years。

1.6 可利用的公用工程条件|Available Utility Conditions

1.6.1 电源|Power Supply

本装置拟采用天然气发电。

Natural gas power generation is proposed for the plant.

高压电源: 10kV $\pm$ 10%、50Hz $\pm$ 10%。

High-voltage power supply: 10kV $\pm$ 10%, 50Hz $\pm$ 10%

低压电源: 400V $\pm$ 10%、50Hz $\pm$ 10%。

Low-voltage power supply: 400V $\pm$ 10%, 50Hz $\pm$ 10%

第 2 章 产品方案和公用工程消耗

Chapter 2 Product Scheme and Utility Consumption

以下数据均基于第 1 章装置的设计基础条件。

The following data are all based on the plant's basic design conditions specified in Chapter 1.

2.1 生产规模和产品方案|Production Scale and Product Scheme

2.1.1 产品方案|Product Proposal

本装置的产品为液化天然气 (LNG) , 产品见表 2.1-1A/B。

The product of this plant is Liquefied Natural Gas (LNG), as presented in Table 2.1-1A/B.

表 2.1-1A LNG 产品

Table 2.1-1A LNG Product

项目 Item	理论计算数据 Theoretical Calculation Data		备注 Remarks
LNG 产量 LNG Output	MMSCFD	4.76	未扣除燃料气 Fuel gas not deducted
	液体状态 m ³ /d Liquid state: m ³ /d: 223	223	组分变化将导致液体体 积和质量的变化 Changes in composition will lead to variations in liquid volume and mass
	质量 ton/d Mass: ton/d: 102.4	102.4	
贮存压力 Storage Pressure	MPa.G	0.2	
贮存温度 Storage Temperature	°C	-160	
LNG 密度 LNG Density	kg/m ³	459.4	

2.2 公用工程消耗|Utility Consumption

装置在上述产品及产量下的日消耗量见表 2.2-1。

The daily consumption of the plant under the above product and output conditions is shown in Table 2.2-1.

表 2.2-1 装置日消耗一览表
Table 2.2-1 Summary of Daily Plant Consumption

消耗项目 Consumption Item	耗 量 Consumption	备 注 Remarks
原料天然气 Feed Natural Gas	5MMSCFD	含燃料气 Including fuel gas
工艺性用电 kW.h Process Power Consumption	48360	按轴功率计算，不含辅机 Calculated by shaft power, excluding auxiliary equipment
每吨 LNG 消耗的电耗 Power Consumption per Ton of LNG	504kw.h/ton LNG	不含公用工程用电 Excluding utility power consumption

2.2.1 装置电耗|Plant Power Consumption

装置在上述产品及产量下的电耗见表 2.2-3。

The power consumption of the main process plant under the above product and output conditions is shown in Table 2.2-3.

表 2.2-3 工艺主装置电耗一览表(kW)
Table 2.2-3 Summary of Power Consumption of Main Process Plant (kW)

项目 Item	功率 (kW) Power(KW)		电压等级 Voltage Grade	备注 Remarks
	轴功 Shaft Power	电机功率 Motor Power		
原料气压缩机 Feed Gas Compressor	145	185	400V、50Hz	活塞机 Reciprocating compressor
贫液泵 Lean Solution Pump	40	55	400V、50Hz	离心泵 Centrifugal pump

项目 Item	功率 (kW) Power(KW)		电压等级 Voltage Grade	备注 Remarks
	轴功 Shaft Power	电机功率 Motor Power		
热端冷剂压缩机 Hot-end Refrigerant Compressor	976	1200	10kV、50Hz	螺杆压缩机 Screw compressor
冷端冷剂压缩机 Cold-end Refrigerant Compressor	880	1000	10kV、50Hz	螺杆压缩机 Screw compressor
各类空冷器用电 Power for Various Air Coolers	100	100	220V、50Hz	
各类控制用电 Power for Various Control Systems	10	10	220V、50Hz	
合 计 Total	2151			

2.2.2 装置冷却方式|Plant Cooling Method

本装置采用空冷。

Air cooling is adopted for the plant.

空冷器设计环境温度 34℃

Design ambient temperature of air coolers: 34℃

空冷器出口介质温度 49℃

Outlet medium temperature of air coolers: 49℃

2.2.3 燃料气消耗|Fuel Gas Consumption

主要为燃气发电机组、火炬长明灯提供燃料气。

Mainly for gas-fired generator sets and torch pilot lights.

燃料气压力：28.82psig

Fuel gas pressure: 28.82psig

燃料气消耗量：~0.18MMSCFD

Fuel gas consumption: ~0.18MMSCFD

2.2.4 装置其他消耗|Other Plant Consumption

装置在上述产品及产量下的其他消耗量见表 2.2-5。

Other consumption of the plant under the above product and output conditions is shown in Table 2.2-5.

表 2.2-5 装置其他消耗一览表

Table 2.2-5 Summary of Other Plant Consumption

消耗项目 Consumption Item	消耗量 Consumption	备注 Remarks
活化 MDEA 胺液 Activated MDEA Amine Solution	~8t	首次用量 Initial dosage
4A 分子筛 4A Molecular Sieve	~6m ³	每 3 年更换一次 Replaced every 3 years
3A 分子筛 3A Molecular Sieve	~0.5m ³	每 3 年更换一次 Replaced every 3 years
浸硫活性炭 Sulfur-impregnated Activated Carbon	~1.5m ³	每 3 年更换一次 Replaced every 3 years
备注：以上用量以最终设计为准。 Note: The above dosages are subject to the final design.		

2.3 三废排放|Waste Discharge

装置在正常运行时的废气、废液和固体废物的排放情况，见表 2.3-1。

The discharge of waste gas, waste liquid and solid waste during the normal operation of the plant is shown in Table 2.3-1.

表 2.3-1 装置三废排放表

Table 2.3-1 Plant Waste Discharge Table

序号 No.	类别 Item	排放点 Discharge Point	排放量 Discharge Capacity	排放规律 Discharge Pattern	组成 Composition	备注 Remarks
一	废气 Waste Gas					
1	酸气 Acid Gas	酸气分离器 Acid Gas Separator	~0.1515MMSCFD	连续 Continuous	CO ₂ +H ₂ O	就地排放 Discharged on-site
二	固废 Solid Waste					
1	3A 分子筛 3A Molecular Sieve	冷剂干燥器 Refrigerant Dryer	~0.5m ³	1 次/3 年 Once every 3 years		
2	4A 分子筛 4A Molecular Sieve	干燥塔 Drying Tower	~6m ³	1 次/3 年 Once every 3 years		
3	浸硫活性炭 Sulfur-impregnated Activated Carbon	脱汞塔 Mercury Removal Tower	~1.5m ³	1 次/3 年 Once every 3 years	活性炭类	
三	废水 Waste Water	分离器 Separator	~13kg/h	连续 Continuous	含烃废水 Hydrocarbon-containing waste water	

第 3 章 工艺技术方案及说明

Chapter 3 Process Technical Scheme and Description

3.1 系统概述|System Overview

天然气液化的工艺过程基本包括原料气预处理（净化）、液化、制冷剂循环压缩、产品储存、装车及辅助系统等，主要工艺流程包括原料气净化、净化气液化工艺。

The process of natural gas liquefaction basically includes feed gas pretreatment (purification), liquefaction, refrigerant cycle compression, product storage, loading and auxiliary systems. The main process flows include feed gas purification and purified gas liquefaction processes.

3.1.1 设计方案概述|Design Scheme Overview

原料天然气经调压、过滤分离、增压、计量后进入天然气预处理系统，脱去 CO₂、H₂O、Hg 和重烃之后进入液化冷箱，在板翅式换热器中被冷却、液化、过冷、节流作为 LNG 产品进入 LNG 储罐。

After pressure regulation, filtration, separation, pressurization and metering, the feed natural gas enters the natural gas pretreatment system to remove CO₂, H₂O, Hg and heavy hydrocarbons, and then enters the liquefaction cold box. In the plate-fin heat exchanger, the gas is cooled, liquefied, subcooled and throttled to become LNG product, which is then sent to the LNG storage tank.

本装置的主要工艺技术方法是：

The main process technologies adopted for the plant are as follows:

- 采用 MDEA 脱除二氧化碳；

MDEA is used to remove carbon dioxide;

- 采用分子筛脱除微量水分；

Molecular sieve is used to remove trace moisture;

■ 采用浸硫活性炭脱除汞；

Sulfur-impregnated activated carbon is used to remove mercury;

■ 采用精密过滤元件过滤分子筛和活性炭粉尘；

Precision filter elements are used to filter molecular sieve and activated carbon dust;

■ 采用 MRC（混合冷剂）循环制冷工艺将净化天然气全部液化；

Mixed Refrigerant Cycle (MRC) refrigeration process is used to fully liquefy the purified natural gas.

本装置主要包括以下系统：

The plant mainly consists of the following systems:

1、调压、过滤分离、增压、计量系统 Pressure Regulation, Filtration & Separation, Pressurization and Metering System	
2、脱碳系统 Decarburization System	
3、干燥及脱汞系统 Drying and Mercury Removal System	
4、液化系统 Liquefaction System	
4.1	液化冷箱（内含铝制板翅式换热器及分离器） Liquefaction Cold Box (including aluminum plate-fin heat exchanger and separator)
4.2	冷剂压缩系统 Refrigerant Compression System
5、产品储存及装车系统 Product Storage and Loading System	
6、BOG 回收系统 BOG Recovery System	

7、辅助系统 Auxiliary Systems	
7.1	PSA 制氮系统 PSA Nitrogen Generation System
7.2	仪表空气系统 Instrument Air System
7.3	混合冷剂配比系统 Mixed Refrigerant Proportioning System
7.4	燃料气系统 Fuel Gas System
7.5	安全泄放系统 Safety Venting System
7.6	污水储存系统 Waste Water Storage System

3.1.2 供货范围概述|Supply Scope Overview

3.1.2.1 乙方供货说明|Supplier's Supply Description

按各工艺系统和成 Set 机组供货为原则。乙方将提供 1Set5MMSCFD 的天然气液化装置。各系统和机组的供货范围除主体设备和单机组外，还包括设备本体所必要的辅助设备、分析仪、仪器仪表、一年备件、管件、阀门、设备内填料，以及相应的底盘、支架、平 Set、成对法兰（带垫片及紧固件）、地脚螺栓等；设备供货原则为撬装的方式。

The supply shall be based on the principle of supplying each process system and complete unit as a whole. The Supplier shall provide one set of 5MMSCFD natural gas liquefaction plant. The supply scope of each system and unit, in addition to the main equipment and single units, shall also include necessary auxiliary equipment, analyzers, instruments and meters, spare parts for one year, pipe fittings, valves, internal fillers of equipment, as well as corresponding chassis, supports,

platforms, paired flanges (with gaskets and fasteners), anchor bolts, etc.; the equipment shall be supplied in a skid-mounted manner.

主要包括以下部分：

The supply mainly includes the following parts:

主工艺配 Set 设备：工艺界区内乙方自制及乙方采购设备；

Main process supporting equipment: equipment manufactured by the Supplier and purchased by the Supplier within the process battery limit;

仪控系统：工艺界区的所有仪控系统；

Instrument and control system: all instrument and control systems within the process battery limit;

阀门：工艺界区的所有阀门（包括设备本体仪表取样根部阀）；

Valves: all valves within the process battery limit (including root valves for instrument sampling on the equipment body);

橇内工艺管道、管件、防腐、保温材料及配 Set 材料等；

Process pipelines, pipe fittings, anti-corrosion and thermal insulation materials and supporting materials on the skids;

试车机物料：分子筛、活性炭（其余物料如胺液、冷剂等由甲方自备）。

Commissioning materials: molecular sieves, activated carbon (other materials such as amine solution and refrigerants shall be prepared by the Client).

3.2 各系统说明|Description of Each System

3.2.1 原料气调压、过滤分离、增压、计量系统|Feed Gas Pressure Regulation, Filtration & Separation, Pressurization and Metering System

1) 工艺描述|Process Description

自上游来的原料天然气经过管道以 402.1psi(28bar)经调压后进入原料气进口过滤分离器，经过分离、增压和计量后进入后续系统。

The feed natural gas from the upstream is transported via pipeline at 402.1psi(28bar), then

enters the feed gas inlet filter separator after pressure regulation, and is sent to the subsequent systems after separation, pressurization and metering.

其主要工艺设备为原料气过滤分离器、原料气压缩机、流量计等。

The main process equipment includes feed gas filter separator, feed gas compressor, flow meter, etc.

2) 设计参数|Design Parameters

原料气进装置流量|Feed gas flow rate into the plant: 5MMSCFD

进装置工作压力|Operating pressure of feed gas into the plant: 402.1psi(28bar)

压缩机排气压力|Compressor discharge pressure: 55bar

压缩机排气温度|Compressor discharge temperature: $\leq 40^{\circ}\text{C}$

压缩机类型|Compressor type: 活塞机|:Reciprocating compressor

压缩机 Set 数|Number of compressors: 2 (1 用 1 备) |2 (1 for use, 1 for standby)

3) 适应范围

50%~110%。

3.2.2 原料气脱酸气单元|Feed Gas Acid Removal Unit

1) 工艺描述|Process Description

从上游来的原料气进入脱酸气单元, 本单元采用 MDEA 溶液的方法脱除原料气中的 CO_2 和 H_2S 等酸性气体。

The feed gas from the upstream enters the deacidification unit, where MDEA solution is used to remove acidic gases such as CO_2 and H_2S from the feed gas.

天然气从吸收塔下部进入, 自下而上通过吸收塔; 完全再生后的 MDEA 溶液 (贫液) 从吸收塔上部进入, 自上而下通过吸收塔, 逆向流动的 MDEA 溶液和天然气在吸收塔内充分接触, 气体中的 CO_2 被吸收而进入液相, 未被吸收的组份从吸收塔顶部引出, 进入脱碳气冷却器和分离器。出脱碳气分离器的气体进入原料气干燥单元, 冷凝液去闪蒸罐。

Natural gas enters the bottom of the absorption tower and flows upward through the tower; the fully regenerated MDEA solution (lean solution) enters the top of the absorption tower and flows downward through the tower. The countercurrently flowing MDEA solution and natural gas are in full contact in the absorption tower, CO₂ in the gas is absorbed into the liquid phase, and the unabsorbed components are led out from the top of the absorption tower and sent to the decarbonized gas cooler and separator. The gas from the decarbonized gas separator enters the feed gas drying unit, and the condensate is sent to the flash tank.

处理后的天然中 CO₂ 含量小于 50ppmV。

The CO₂ content in the treated natural gas is less than 50ppmV.

吸收了 CO₂ 的 MDEA 溶液称富液降压后与再生塔底部流出的溶液（贫液）换热后，升温到 ~95°C 去再生塔上部，在再生塔进行汽提再生，直至贫液的贫液度达到指标。

The MDEA solution that has absorbed CO₂ (rich solution) is depressurized, heat-exchanged with the solution (lean solution) flowing out from the bottom of the regeneration tower, heated to ~95°C and sent to the top of the regeneration tower for stripping and regeneration until the lean degree of the lean solution meets the index.

出再生塔的贫液经过贫富液换热器、贫液冷却器，贫液被冷却到 ~40°C，被贫液泵加压后，从吸收塔上部进入。

The lean solution from the regeneration tower is cooled to ~40°C after passing through the rich-lean solution heat exchanger and lean solution cooler, pressurized by the lean solution pump, and then enters the top of the absorption tower.

再生塔顶部出口气体经酸气冷却器，进入酸气分离器，出酸气分离器的气体送往酸气排放系统，冷凝液返回 MDEA 缓冲罐。

The gas from the top of the regeneration tower passes through the acid gas cooler and enters the acid gas separator. The gas from the acid gas separator is sent to the acid gas discharge system, and the condensate is returned to the MDEA buffer tank.

再生塔再沸器的热源由导热油加热。

The heat source for the reboiler of the regeneration tower is provided by heat transfer oil

heating.

本单元主要工艺设备为吸收塔和再生塔。

The main process equipment of this unit includes absorption tower and regeneration tower.

2) 设计参数|Design Parameters

处理量 Processing capacity	5MMSCFD
原料气中 CO ₂ 含量设计点 Design CO ₂ content in feed gas	3% (mol%)
吸收塔操作压力 Absorption tower operating pressure	55bar
吸收塔操作温度 Absorption tower operating temperature	40~49°C
再生塔操作压力 Regeneration tower operating pressure	0.2~0.4barg
再生塔操作温度 Regeneration tower operating temperature	95~120°C
再生热源 Regeneration heat source	导热油加热
脱碳气中 CO ₂ 气体的含量 CO ₂ content in decarbonized gas	≤50ppm
脱碳气中 H ₂ S 气体的含量 H ₂ S content in decarbonized gas	≤4ppm

3) 适应范围|Applicable Range

50%~110%。

3.2.3 原料气干燥及脱汞单元|Feed Gas Drying and Mercury Removal Unit

1) 工艺描述|Process Description

本装置采用变温吸附技术进行气体分离提纯，变温吸附技术是以吸附剂（多孔固体物质）内部表面对气体分子的物理吸附为基础，利用吸附剂对气体的吸附容量随吸附温度和压力不同而变化的特性，吸附剂对不同气体组份有选择性吸附的条件下，低温高压时吸附混合气中的某些组份，未被吸附组份通过吸附器层流出，高温高压时脱附这些被吸附的组份，以进行下一次低温高压吸附，可采用多个吸附塔而达到气体的连续分离的目的。

The plant adopts temperature swing adsorption (TSA) technology for gas separation and

purification. TSA is based on the physical adsorption of gas molecules on the internal surface of adsorbents (porous solid materials). Utilizing the characteristic that the adsorption capacity of adsorbents for gases changes with adsorption temperature and pressure, and under the condition that adsorbents have selective adsorption for different gas components, certain components in the mixed gas are adsorbed at low temperature and high pressure, the unadsorbed components flow out through the adsorber layer, and these adsorbed components are desorbed at high temperature and high pressure for the next adsorption at low temperature and high pressure. Multiple adsorption towers can be used to achieve continuous gas separation.

原料气干燥单元设三 Set 吸附器切换操作, 其中一 Set 吸附、一 Set 冷吹、一 Set 再生。

Three adsorbers are set up in the feed gas drying unit for switch operation, among which one is for adsorption, one for cold blowing, and one for regeneration.

从原料天然气脱酸性气体单元来的原料气进入吸附器顶部, 通过分子筛吸附脱除水分后, 从吸附器底部出来, 脱除水后天然气中进入原料气脱汞及粉尘过滤单元。

The feed gas from the feed natural gas deacidification unit enters the top of the adsorber, the moisture is removed by molecular sieve adsorption, and the gas flows out from the bottom of the adsorber and enters the feed gas mercury removal and dust filtration unit.

原料气干燥单元用少量的原料气作为冷吹和再生介质, 再生气出吸附塔后通过冷却、分离后返回吸附塔入口。

A small amount of feed gas is used as the cold blowing and regeneration medium in the feed gas drying unit. After cooling and separation, the regeneration gas from the adsorber is returned to the inlet of the adsorber.

再生气首先从上而下通过冷却状态的吸附器, 之后再生气通过再生加热器加热至再生温度 220~260°C, 然后从吸附器底部进入, 将吸附剂吸附的水解吸。再生气从干燥器顶部出来。

The regeneration gas first flows downward through the cooled adsorber, then is heated to the regeneration temperature of 220~260°C by the regeneration heater, and then enters from the bottom of the adsorber to desorb the water adsorbed by the adsorbent. The regeneration gas flows out from the top of the dryer.

来自吸附器后的天然气进入浸硫活性炭吸附器，汞与浸硫活性炭上的硫产生化学反应生成硫化汞，吸附在活性炭上，从而达到脱除汞之目的。从脱汞器出来的天然气的汞含量小于 $0.01\mu\text{g}/\text{Nm}^3$ 。

The natural gas from the adsorber enters the sulfur-impregnated activated carbon adsorber, where mercury reacts chemically with sulfur on the sulfur-impregnated activated carbon to form mercury sulfide, which is adsorbed on the activated carbon, thereby achieving mercury removal. The mercury content in the natural gas from the mercury removal tower is less than $0.01\mu\text{g}/\text{Nm}^3$.

脱汞器设置一 Set，浸硫活性炭按照检测情况更换。

One mercury removal tower is installed, and the sulfur-impregnated activated carbon is replaced according to the detection results.

脱水脱汞后设两 Set 过滤器，根据阻力数据切换使用，达到过滤分子筛之目的。经脱粉尘后，原料气中的粉尘颗粒小于 $5\mu\text{m}$ 。

Two filters are installed after dehydration and mercury removal, which are switched according to resistance data to filter molecular sieves. After dust removal, the dust particles in the feed gas are less than $5\mu\text{m}$.

经过该单元后，干燥天然气中的水 $\leq 1\text{ppm}$ 。

After passing through this unit, the water content in the dried natural gas is $\leq 1\text{ppm}$.

主要设备为吸附塔、再生加热器、再生气冷却器和再生气分离器等。

The main equipment includes adsorber, regeneration heater, regeneration gas cooler, regeneration gas separator, etc.

2) 设计参数|Design Parameters

原料气处理量 Feed gas processing capacity	5MMSCFD
-------------------------------------	---------

操作压力 Operating pressure	55bar
-------------------------	-------

吸附温度 Adsorption temperature	49°C
-----------------------------	------

原料气含水量 Feed gas water content	在进气条件下饱和 aturated under inlet conditions
再生方式 Regeneration method	等压再生 Isobaric regeneration
再生温度 Regeneration temperature	220~260°C
再生热源 Regeneration heat source	导热油加热 Heat transfer oil heating
净化气中 H ₂ O 的含量 H ₂ O content in purified gas	≤1ppm
过滤精度 Filtration precision	≤5μm
3) 适应范围 Applicable Range	
50%~110%。	

3.2.4 液化单元|Liquefaction Unit

1) 工艺描述|Process Description

来自原料气干燥及脱汞单元后的原料气一部分做燃料气进入燃料气单元，其余部分进入液化单元。

A part of the feed gas from the feed gas drying and mercury removal unit is used as fuel gas and enters the fuel gas unit, and the rest enters the liquefaction unit.

净化后的天然气首先进入冷箱内的液化换热器被返流的低温混合冷剂逐渐冷却到-162°C后节流得到 LNG 产品。

The purified natural gas first enters the liquefaction heat exchanger in the cold box, is gradually cooled to -162°C by the countercurrent low-temperature mixed refrigerant, and then is throttled to obtain LNG product.

2) 设计参数|Design Parameters

原料气处理量 Feed gas processing capacity	4.963MMSCFD
操作压力 Operating pressure	55bar

液化温度|Liquefaction temperature -162°C

3) 适应范围|Applicable Range

50%~110%。

3.2.5 制冷单元|Refrigeration Unit

本装置的制冷包括混合制冷单元，其中混合制冷单元主要为天然气液化提供冷量。

The refrigeration of the plant includes a mixed refrigerant unit, which mainly provides cooling capacity for natural gas liquefaction.

1) 工艺描述|Process Description

本装置的混合冷剂将由净化的甲烷、氮气和烃类冷剂组成。

The mixed refrigerant of the plant consists of purified methane, nitrogen and hydrocarbon refrigerants.

主要设备为冷剂循环压缩机、冷剂配比罐等。

The main equipment includes refrigerant cycle compressor, refrigerant proportioning tank, etc.

2) 设计参数|Design Parameters

混合制冷压缩机类型 螺杆压缩机

Mixed refrigerant compressor type: Screw compressor

压缩机 Set 数 2Set (高低温压缩机各 1Set)

Number of compressors: 2 (1 for high temperature, 1 for low temperature)

吸入压力 0.4MPa.G

Suction pressure: 0.4MPa.G

吸入温度 46.5°C

Suction temperature: 46.5°C

排气压力 1.9MPa.G (指末级冷却器出口法兰处)

Discharge pressure: 1.9MPa.G (referring to the flange at the outlet of the final stage cooler)

排气温度 ≤40°C (指末级冷却器出口法兰处)

Discharge temperature: $\leq 40^{\circ}\text{C}$ (referring to the flange at the outlet of the final stage cooler)

3) 适应范围|Applicable Range

50%~110%。

3.2.6 冷剂储存、配比及回收单元|Refrigerant Storage, Proportioning and Recovery Unit

1) 工艺描述|Process Description

除甲烷（净化后的天然气）和氮气两种冷剂外，其余三种冷剂均有各自的储罐提供。当系统在停车检修时，为避免 MR 制冷单元的冷剂排放而造成的浪费，装置设置了冷剂回收系统，停车时，首先将冷剂压缩机级间分离器内的冷剂液体由管道输送到冷剂回收罐，再将冷剂压缩机出口分离器中的液体排入冷剂回收罐。当系统临时停车或非冷剂系统原因而停车时，因设备和管道在设计时充分考虑了冷剂因挥发压力要升高的因素，在冷剂系统设计时提高了设计压力。因此，一般情况下，制冷系统无需释放压力。如确因系统压力过高造成压缩机启动困难，则可适当排放少量气体到火炬即可。

Except for two refrigerants (methane and nitrogen), the other three refrigerants are supplied by their respective storage tanks. To avoid waste caused by the discharge of refrigerant from the MR refrigeration unit during plant shutdown and maintenance, a refrigerant recovery system is installed in the plant. During shutdown, the refrigerant liquid in the interstage separator of the refrigerant compressor is first transported to the refrigerant recovery tank through pipelines, and then the liquid in the outlet separator of the refrigerant compressor is discharged into the refrigerant recovery tank. In case of temporary plant shutdown or shutdown due to non-refrigerant system reasons, the design pressure of the refrigerant system is increased in the design, as the equipment and pipelines have fully considered the factor of refrigerant pressure rise due to volatilization. Therefore, generally, there is no need to release pressure from the refrigeration system. If the compressor is difficult to start due to excessively high system pressure, a small amount of gas can be appropriately discharged to the torch.

a) 甲烷（净化天然气）的配比|Methane (purified natural gas) proportioning

制冷系统中的甲烷来自净化后的天然气,在净化后的天然气进冷箱前通过一分支经调压、计量后进入冷剂配比罐;

Methane in the refrigeration system is derived from purified natural gas. Before the purified natural gas enters the cold box, a branch is taken, and the methane is regulated in pressure, metered and then sent to the refrigerant proportioning tank.

b) 氮气的配比|Nitrogen proportioning

制冷系统中的氮气可由液氮储罐汽化复热后提供,也可由装置的 PSA 提供,来自系统的氮气经调压、计量后进入冷剂配比罐。

Nitrogen in the refrigeration system can be provided by vaporization and reheating of liquid nitrogen storage tank, or by the PSA of the plant. Nitrogen from the system is regulated in pressure, metered and then sent to the refrigerant proportioning tank.

c) 乙烯储存与配比|Ethylene storage and proportioning

通过乙烯槽车运输来的乙烯,卸入乙烯储罐,储罐内的液态乙烯经汽化器汽化复热后再经调压、计量后进入冷剂配比罐。

Ethylene transported by ethylene tank truck is unloaded into the ethylene storage tank. The liquid ethylene in the storage tank is vaporized and reheated by the vaporizer, then regulated in pressure, metered and sent to the refrigerant proportioning tank.

d) 丙烷储存与配比|Propane storage and proportioning

通过丙烷槽车运输来的丙烷,卸入丙烷储罐,储罐内的液态丙烷经减压汽化后计量进入冷剂配比罐。

Propane transported by propane tank truck is unloaded into the propane storage tank. The liquid propane in the storage tank is vaporized under reduced pressure, metered and then sent to the refrigerant proportioning tank.

e) 异丁烷储存与配比|Isobutane storage and proportioning

通过异丁烷槽车运输来的异丁烷,卸入异丁烷储罐,储罐内的液态异丁烷经减压汽化后

计量进入冷剂配比罐。

Isobutane transported by isobutane tank truck is unloaded into the isobutane storage tank. The liquid isobutane in the storage tank is vaporized under reduced pressure, metered and then sent to the refrigerant proportioning tank.

2) 设计参数|Design Parameters

液氮储罐 Liquid nitrogen storage tank	20m ³
乙烯储罐 Ethylene storage tank	20m ³
异丁烷储罐 Isobutane storage tank	10m ³
丙烷储罐 Propane storage tank	10m ³
高温冷剂回收罐 High-temperature refrigerant recovery tank	10m ³

3.2.7 系统干燥和解冻单元|System Drying and Thawing Unit

1) 工艺描述|Process Description

a) 干燥|Drying

初次开车之前，需对分子筛干燥单元下游的天然气管路和冷剂制冷管路全部进行干燥，此时采用将来自于再生气加热器出口的热气体和分子筛干燥塔出口的冷气体混合，进入到系统中进行干燥。

Before the initial start-up, all natural gas pipelines and refrigerant refrigeration pipelines downstream of the molecular sieve drying unit must be dried. At this time, the hot gas from the outlet of the regeneration heater and the cold gas from the outlet of the molecular sieve drying tower are mixed and sent into the system for drying.

B) 解冻|Thawing

当冷箱因水、二氧化碳或重烃在低温情况下凝固而堵塞管道的时候，需利用加温氮气或天然气对液化冷箱进行加温解冻。此时需将由氮气管网来的氮气经加热器加热到~65℃后进入液化冷箱的原料气正反流通道、制冷剂正反流通道进行加热，之后从冷箱顶部排出。

When the cold box pipeline is blocked due to solidification of water, carbon dioxide or heavy hydrocarbons at low temperature, the liquefaction cold box needs to be heated and thawed by heated nitrogen or natural gas. At this time, nitrogen from the nitrogen pipeline network is heated to $\sim 65^{\circ}\text{C}$ by the heater, then sent into the forward and reverse flow channels of feed gas and refrigerant of the liquefaction cold box for heating, and then discharged from the top of the cold box.

2) 设计参数|Design Parameters

加温解冻的天然气或氮气温度|Temperature of natural gas or nitrogen for heating and thawing: $\sim 65^{\circ}\text{C}$

加温解冻温升速率|Temperature rise rate for heating and thawing: $\leq 40^{\circ}\text{C/h}$

加温解冻时间|Heating and thawing time: 24~48h

3.2.8 产品储存及装车单元|Product Storage and Loading Unit

由液化冷箱来的 LNG 进入 LNG 储存系统，并经 LNG 装车鹤管送至 LNG 槽车外销。

The LNG from the liquefaction cold box enters the LNG storage system, and is then sent to LNG tank trucks for external sales through LNG loading arms.

装置产生的重烃进入重烃储存系统，并经重烃装车鹤管送至重烃槽车外销。

The heavy hydrocarbons produced by the plant enter the heavy hydrocarbon storage system, and are then sent to heavy hydrocarbon tank trucks for external sales through heavy hydrocarbon loading arms.

3.2.9 BOG 回收单元|BOG Recovery Unit

1) 系统功能|System Function

装置的 BOG 主要由储存及装车过程而产生的蒸发气，为节约生产成本将这些 BOG 回收做燃料气。

The BOG of the plant is mainly the boil-off gas generated during storage and loading processes. To save production costs, the BOG is recovered as fuel gas.

2) 设计参数|Design Parameters

BOG 处理量|BOG processing capacity 200Nm³/h

3) 适应范围|Applicable Range

负荷调节范围为 50~100%。

Load adjustment range: 50~100%

3.2.10 污水储存单元|Waste Water Storage Unit

1) 工艺描述|Process Description

装置设置 1Set20m³ 的污水储存系统，装置的污水由槽车外运处理。

One set of 20m³ waste water storage system is installed in the plant, and the waste water of the plant is transported out for treatment by tank truck.

2) 设计参数|Design Parameters

容积：20m³

Volume: 20m³

设计压力：0.2MPa

Design pressure: 0.2MPa

3.2.11 燃料气单元|Fuel Gas Unit

1) 系统功能|System Function

主要为燃气发电机组、火炬长明灯提供燃料气。

Mainly to provide fuel gas for gas-fired generator sets and torch pilot lights.

2) 工艺流程描述|Process Flow Description

燃料气主要有 BOG。

The fuel gas is mainly BOG.

在燃料气量不足或开机前调试时，由进入装置的原料气补充。

When the fuel gas supply is insufficient or during commissioning before start-up, it is

supplemented by the feed gas entering the plant.

3.2.12 安全排放单元|Safety Venting Unit

1) 系统功能|System Function

用来处理各类可燃气体正常、事故、开停车及检修时的放空。火炬系统保证生产装置正常或事故排放时能够及时、安全、可靠地在筒体中排放燃烧，并满足国家现行相关标准和环保要求。

Used to handle the venting of various flammable gases during normal operation, accidents, start-up, shutdown and maintenance. The torch system ensures that the flammable gases can be discharged and burned in the cylinder in a timely, safe and reliable manner during normal or accident discharge of the production plant, and meets the current relevant national standards and environmental protection requirements.

2) 设计参数|Design Parameters

详细设计参数在工程设计时确定。

The detailed design parameters shall be determined during engineering design.

第 4 章 设备选型和供货

Chapter 4 Equipment Selection and Supply

本液化装置所供设备按分单元撬块化设计。各撬块设备除主体设备、一年正常使用的备件、螺杆压缩机专用工具、仪表和阀门外，还包括撬块上所有接管的成对法兰（带垫片及紧固件）、地脚螺栓等。

The equipment supplied for this liquefaction plant is designed in a skid-mounted manner by unit. In addition to the main equipment, spare parts for normal use for one year, special tools for screw compressors, instruments and valves, each skid-mounted equipment shall also include paired flanges (with gaskets and fasteners) for all nozzles on the skid, anchor bolts, etc.

撬块上的电气接口均设置端子箱，同时采用集成技术，最大限度地减少连接电缆数量。

Terminal boxes are installed at all electrical interfaces on the skids, and integration technology is adopted to minimize the number of connecting cables.

撬块设置仪表气总管，内部的仪表气管连接已经在撬块制造时完成。

A main instrument air pipe is installed on the skids, and the connection of internal instrument air pipes is completed during skid manufacturing.

本章所述的工艺设备基本技术数据以最终设计为准。

The basic technical data of the process equipment described in this chapter is subject to the final design.

4.1 工艺设备|Process Equipment

本装置的工艺设备供货清单如下（外形尺寸以最终供货为准）：

The list of process equipment supplied for the plant is as follows (overall dimensions are subject to the final supply):

(1) 原料气过滤分离、计量&燃料气单元

Feed Gas Filtration & Separation, Metering & Fuel Gas Unit

序号 No.	设备 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	备注 Remarks
一	过滤分离及燃料气撬 Filtration, Separation & Fuel Gas Skid		1Set	四川恒重 Sichuan Hengzhong	
1	原料气过滤分离器 Feed Gas Filter Separator	Q345R	1Set	四川恒重 Sichuan Hengzhong	内含滤芯 Including filter element
2	压缩机出口分离器 Compressor Outlet Separator	Q345R	1Set	四川恒重 Sichuan Hengzhong	
3	燃料气缓冲罐 Fuel Gas Buffer Tank	Q345R	1Set	四川恒重 Sichuan Hengzhong	
4	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
5	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set	四川恒重 Sichuan Hengzhong	
二	原料气压缩机组 Feed Gas Compressor Unit		2Set	国产优质 Domestic high-quality	活塞机, 1 用 1 备 Reciprocating compressor, 1 for use, 1 for standby

(2) 原料气脱酸气单元|Feed Gas Acid Removal Unit

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Note
------------	-------------------	----------------	-----------	--------------------	--------------

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Note
三	脱碳撬 (一) Decarburization Skid (1)		1Set	四川恒重 Sichuan Hengzhong	
1	吸收塔 Absorption Tower	Q345R	1Set	四川恒重 Sichuan Hengzhong	内含不锈钢规整填料 Including stainless steel structured packing
2	再生塔 Regeneration Tower	S30408	1Set	四川恒重 Sichuan Hengzhong	内含不锈钢规整填料 Including stainless steel structured packing
3	再沸器 Reboiler	S30408	1Set	四川恒重 Sichuan Hengzhong	管壳式 Shell and tube type
4	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
5	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set	四川恒重 Sichuan Hengzhong	
四	脱碳撬 (二) Decarburization Skid (2)		1Set	四川恒重 Sichuan Hengzhong	
1	脱酸气冷却器 Acid Removal Gas Cooler		1Set	国产优质 Domestic high-quality	空冷器 Air cooler

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Note
2	脱酸气分离器 Acid Removal Gas Separator	Q345R	1Set	四川恒重 Sichuan Hengzhong	
3	MDEA 循环泵 MDEA Circulation Pump		2Set	国产优质 Domestic high-quality	用 1 备 1, 离心式 1 for use, 1 for standby, centrifugal type
4	贫富液换热器 Rich-Lean Solution Heat Exchanger	S30408	1Set	上海浩特等	板片式 Plate type
5	贫液冷却器 Lean Solution Cooler		1Set	国产优质 Domestic high-quality	空冷器 Air cooler
6	活性炭过滤器 Activated Carbon Filter	S30408	1Set	四川恒重 Sichuan Hengzhong	
7	机械过滤器 Mechanical Filter	S30408	1Set	四川恒重 Sichuan Hengzhong	内含滤芯 Including filter element
8	酸气冷却器 Acid Gas Cooler	S30408	1Set	国产优质 Domestic high-quality	空冷器 Air cooler
9	酸气分离器 Acid Gas Separator	S30408	1Set	四川恒重 Sichuan Hengzhong	
10	MDEA 循环槽	Q345R	1Set	四川恒重	

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Note
	MDEA Circulation Tank			Sichuan Hengzhong	
11	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
12	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set	四川恒重 Sichuan Hengzhong	

(3) 原料气干燥及脱汞单元|Feed Gas Drying and Mercury Removal Unit

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
五	原料气干燥、脱汞撬 Feed Gas Drying & Mercury Removal Skid		1Set	四川恒重 Sichuan Hengzhong	
1	干燥塔 Drying Tower	Q345R	3Set	四川恒重 Sichuan Hengzhong	内含分子筛 Including molecular sieve
2	再生气加热器 Regeneration Gas Heater	Q345R	1Set	四川恒重 Sichuan Hengzhong	管壳式 Shell and tube type
3	再生气冷却器 Regeneration Gas Cooler	20	1Set	国产优质 Domestic high-quality	空冷器 Air cooler
4	再生气分离器	Q345R	1Set	四川恒重 Sichuan	

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
	Regeneration Gas Separator			Hengzhong	
5	粉尘过滤器 Dust Filter	Q345R	2Set	四川恒重 Sichuan Hengzhong	内含滤芯 Including filter element
6	脱汞塔 Mercury Removal Tower	Q245R	1Set	四川恒重 Sichuan Hengzhong	内含脱汞剂 Including mercury removal agent
7	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
8	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set	四川恒重 Sichuan Hengzhong	

(4) 低温液化单元|Low-Temperature Liquefaction Unit

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
七	液化冷箱 Liquefaction Cold Box		1Set	苏州三川等 Suzhou Sanchuan, etc.	
1	液化换热器 Liquefaction Heat Exchanger		1 组	苏州三川等 Suzhou Sanchuan, etc.	铝制板翅式换热器 Aluminum plate-fin heat exchanger
2	重烃分离器 Heavy Hydrocarbon Separator		1Set		

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
3	调节阀门 Regulating Valves		1Set		
4	一次仪表 Primary Instruments		1Set		
5	低温管道和手动阀门 Low-Temperature Pipelines and Manual Valves		1Set		
6	冷箱绝热材料 Cold Box Thermal Insulation Material		足量		
7	冷箱壳体 Cold Box Shell		1Set		
八	残液加热器 Residue Heater		1Set	国产 Domestic	空温式 Air temperature type
九	氮气电加热器 (20KW) Nitrogen Electric Heater (20KW)		1Set	国产 Domestic	

(6) MR 循环制冷单元|MR Cycle Refrigeration Unit

序 号 No.	设备名称 Equipment		数量 Qty	厂家 Manufacturer	其他说明
十	热端冷剂压缩机 Hot-end Refrigerant Compressor		1Set	国产 Domestic	螺杆机, 空冷 Screw compressor, air-cooled
十一	冷端冷剂压缩机 Cold-end Refrigerant Compressor		1Set	国产 Domestic	螺杆机, 空冷 Screw compressor, air-cooled

(7) 冷剂储存、配比及回收单元|Refrigerant Storage, Proportioning and Recovery Unit

序 号 No.	设备名称 Equipment		数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
十二	乙烯储存配比撬 Ethylene Storage & Proportioning Skid		1Set	四川恒重 Sichuan Hengzhong	
1	乙烯储罐 (20m ³) Ethylene Storage Tank (20m ³)		1Set	国产	
2	乙烯汽化器 Ethylene Vaporizer		1Set	国产	Flow rate: 300Nm ³ /h
3	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
4	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set		
5	乙烯卸车鹤管 Ethylene Unloading Arm		1Set	国产	Out of skid
十三	液氮储存配比撬 Liquid Nitrogen Storage & Proportioning Skid		1Set	四川恒重 Sichuan Hengzhong	
1	液氮储罐 Liquid Nitrogen Storage Tank		1Set	国产	Capacity: 20m ³
2	液氮汽化器 Liquid Nitrogen Vaporizer		1Set	国产	Flow rate: 300Nm ³ /h
3	阀门、仪表、液位计 Valves, Instruments, Level		1Set		

序 号 No.	设备名称 Equipment		数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
	Gauges				
4	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set		
十四	丙烷储存配比撬 Propane Storage & Proportioning Skid		1Set	四川恒重 Sichuan Hengzhong	
1	丙烷储罐 Propane Storage Tank	Q345R	1Set	四川恒重 Sichuan Hengzhong	Capacity: 10m ³
2	丙烷卸车泵 Propane Unloading Pump		1Set	国产	
3	丙烷干燥器 Propane Dryer	Q345R	1Set	四川恒重 Sichuan Hengzhong	
4	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
5	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set		
6	丙烷卸车鹤管 Propane Unloading Arm		1Set	国产	Out of skid
十五	异丁烷储存配比撬 Isobutane Storage &		1Set	四川恒重 Sichuan	

序 号 No.	设备名称 Equipment		数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
	Proportioning Skid			Hengzhong	
1	异丁烷储罐 Isobutane Storage Tank	Q345R	1Set	四川恒重 Sichuan Hengzhong	Capacity: 10m ³
2	异丁烷卸车泵 Isobutane Unloading Pump		1Set	国产	
3	异丁烷干燥器 Isobutane Dryer	Q345R	1Set	四川恒重 Sichuan Hengzhong	
4	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
5	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set		
6	异丁烷卸车鹤管 Isobutane Unloading Arm		1Set	国产	Out of skid
十六	冷剂回收撬 Refrigerant Recovery Skid		1Set	四川恒重 Sichuan Hengzhong	
1	冷剂回收罐 (热端) Refrigerant Recovery Tank (Hot-end)	Q345R	1Set	四川恒重 Sichuan Hengzhong	

序 号 No.	设备名称 Equipment		数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
2	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
3	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set		

(8) 产品储存装车及 BOG 回收单元 Product Storage, Loading and BOG Recovery Unit

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
十七	BOG 加热撬 BOG Heating Skid		1Set	四川恒重 Sichuan Hengzhong	
1	BOG 空温式复热器 BOG Air Temperature Type Reheater		2Set	国产优质 Domestic high-quality	200Nm ³ /h
2	阀门、仪表、液位计 Valves, Instruments, Level Gauges		1Set		
3	撬块底座(含保温、燃气探头等) Skid Base (including thermal insulation, gas probe, etc.)		1Set	四川恒重 Sichuan Hengzhong	
十八	LNG 真空储罐 LNG Vacuum Storage Tank		5Set	苏州杜尔等	Capacity: 100m³
十九	LNG 装车泵撬 LNG Loading Pump Skid		1Set	四川恒重 Sichuan	

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty	厂家 Manufacturer	其他说明 Other Remarks
				Hengzhong	
1	LNG 装车潜液泵 LNG Loading Submerged Pump		2Set	国产优质 Domestic high-quality	Flow rate: 80m³/h
2	阀门、仪表等 Valves, Instruments, etc.		1Set		
3	撬块底座(含保冷、燃气探头等) Skid Base (including cold insulation, gas probe, etc.)		1Set	四川恒重 Sichuan Hengzhong	
二十	LNG 装车臂 LNG Loading Arm		3Set	国产优质 Domestic high-quality	

(9) 污水储存单元|Waste Water Storage Unit

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty.	厂家 Manufacturer	其他说明 Note
二十四	污水储罐 (20m³/0.2MPa) Waste Water Storage Tank (20m³/0.2MPa)	Q345R	1Set	四川恒重 Sichuan Hengzhong	
二十五	污水装车泵 Waste Water Loading Pump		2Set	国产优质 Domestic high-quality	Flow rate: 15m³/h
二十六	污水装车鹤管 Waste Water Loading Arm		1Set	国产优质 Domestic high-quality	

(10) 手动阀门和过滤器|Manual Valves and Filters

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty.
手动阀门和过滤器 Manual valve and Filter			
1	手动低温阀 Manual Low-Temperature Valves	1Set	成空阀门、四川空分等 Chengdu Air Separation Valve, Sichuan Air Separation, etc.
2	手动中常温阀 Manual Medium and Normal Temperature Valves	1Set	环球阀门、方正阀门集团等 Universal Valve, Fangzheng Valve Group, etc.
3	常温安全阀 Normal Temperature Safety Valves	1Set	罗浮、永一等 Luofu, Yongyi, etc.
4	低温安全阀 Low-Temperature Safety Valves	1Set	成空阀门、四川空分等 Chengdu Air Separation Valve, Sichuan Air Separation, etc.
4	管道过滤器 Pipeline Filters	1Set	环球阀门、方正阀门集团等 Universal Valve, Fangzheng Valve Group, etc.

(11) 公用工程系统|Utility Systems

序 号 No.	设备名称 Equipment	材质 Material	数量 Qty.
仪表空气和氮气系统 Instrument air and nitrogen system			
1	PSA 制氮系统, 80Nm ³ /h PSA Nitrogen Generation System, 80Nm ³ /h	1Set	其中仪表空气为 200Nm ³ /h Instrument air: 200Nm ³ /h; Domestic high-quality 国产优质 Domestic high-quality

说明：主要包括空压机（1 用 1 备）、净化装置、空气储罐、氧氮分离装置、氮气缓冲罐、仪表空气缓冲罐及管路附件等。

Note: Mainly including air compressors (1 for use, 1 for standby), purification device, air storage tank, oxygen-nitrogen separation device, nitrogen buffer tank, instrument air buffer tank and pipeline accessories, etc.

脱盐水系统

Demineralized Water System

1	脱盐系统 Demineralized Water System	1Set	0.5t/h	宏森环保、成都彰华或 同等 Hongsen Environmental Protection, Chengdu Zhanghua or equivalent
---	------------------------------------	------	--------	--

导热油炉

Hot oil

1	余热回收导热油炉(双温型) Heat Transfer Oil Boiler (Dual-temperature Type)	1Set	总出热量：1400kW Total heat output: 1400kW	国产 Domestic
---	--	------	---	----------------

安全泄放单元

Safety Relief Unit

1	安全泄放单元（高架火炬） Safety Relief Unit (Elevated Torch)	1Set		陕西万科等 Shaanxi Wanke, etc.
---	--	------	--	------------------------------

(12) 仪表控制|Instrument and Control

序 号 NO	设备名称 Equipment	数量 Qty	厂家 Manufacturer
一	DCS 系统 DCS System	1Set	和利时、浙江中控 Holleysys, Zhejiang SUPCON
二	SIS 系统	1Set	和利时、浙江中控

	SIS System		Holleysys, Zhejiang SUPCON
三	GDS 系统 GDS System	1Set	海湾、深圳特安、西安博康 Gulf, Shenzhen Techann, Xi'an Bokang
四	控制阀 Control Valves	1Set	含 SIS 系统所需阀门、仪表 Including valves and instruments required for SIS system
1	紧急切断阀（常温） Emergency Shut-off Valves (Normal Temperature)	1Set	富阳恒途、浙江永盛、杭州佳能 Fuyang Hengtu, Zhejiang Yongsheng, Hangzhou Canon
2	分子筛切换阀 Molecular Sieve Switching Valves	1Set	富阳恒途、成空阀门、杭州佳能 Fuyang Hengtu, Zhejiang Yongsheng, Hangzhou Canon
3	其他开关阀 Other On-off Valves	1Set	富阳恒途、浙江永盛、杭州佳能 Fuyang Hengtu, Zhejiang Yongsheng, Hangzhou Canon
4	冷箱调节阀 Cold Box Regulating Valves	1Set	富阳恒途、浙江永盛、杭州佳能 Fuyang Hengtu, Zhejiang Yongsheng, Hangzhou Canon
5	其他调节阀 Other Regulating Valves	1Set	富阳恒途、浙江永盛、杭州佳能 Fuyang Hengtu, Zhejiang Yongsheng, Hangzhou Canon
6	压力、差压变送器 Pressure and Differential Pressure Transmitters	1Set	广州派晟、上润、通博、程铭或同等 Guangzhou PaiSheng, Shangrun, Tongbo, Chengming or equivalent
7	铂热电阻 Platinum Thermal Resistance	1Set	广州派晟、上润、通博、程铭或同等 Guangzhou PaiSheng, Shangrun, Tongbo, Chengming or equivalent
8	标准孔板 Standard Orifice Plate	1Set	广州派晟、上润、通博或同等 Guangzhou PaiSheng, Shangrun, Tongbo, Chengming or equivalent

9	气体涡轮、涡街流量计 Gas Turbine and Vortex Flow Meters	1Set	苍南/天信或同等 Cangnan/Tianxin or equivalent
10	就地液位计 Local Level Gauges	1Set	通博、惠科达、上海星申或同等 Tongbo, Huikeda, Shanghai Xingshen or equivalent
11	不锈钢压力表、差压表 Stainless Steel Pressure Gauges and Differential Pressure Gauges	1Set	天府仪表、布莱迪、川仪或同等 Tianfu Instrument, Brady, Chuanyi or equivalent
12	UPS	1Set	山特/金武士或同等 Santak/Kingchai or equivalent
五	分析系统 Analysis System	1Set	
1	红外 CO ₂ 分析仪 Infrared CO ₂ Analyzer	1Set	西安泰戈等 Xi'an Taige, etc.
2	氧化铝露点分析仪 Alumina Dew Point Analyzer	1Set	西安泰戈等 Xi'an Taige, etc.
3	在线气相色谱分析仪 On-line Gas Chromatograph	1Set	SIEMENS 或 ABB (冷剂分析用) SIEMENS or ABB (for refrigerant analysis)
六	手工分析系统 Manual Analysis System	1Set	
1	纯化水水质分析仪 Purified Water Quality Analyzer	1Set	
2	脱盐水水质分析 Demineralized Water Quality Analysis	1Set	

3	MDEA 溶液分析 MDEA Solution Analysis	1Set	
---	--	------	--

(11) 电气控制|Electrical Control

序号 No.	名称 Item	数量 Qty	厂家 Manufacturer	备注 Note.
1	电气控制系统 Electrical Control System	1Set	四川恒重 Sichuan Hengzhong	